

AI Assignment - 2

1a) Define a constraint satisfaction problem with map colouring problem?

A) constraint satisfaction problem (CSP)

* A constraint satisfaction problem is a problem in Artificial Intelligence where we need to assign values to a set of variables such that all given constraints are satisfied.

components of CSP :-

Variables :- elements that need values.

Domains :- possible values for each variable.

Constraints :- rules restricting the values.

Example :- Map colouring problem

* In map colouring problem, each region on map is a variable and colours are domain.

* The constraint is that no two adjacent regions should have the same colour.

Ex:-

Variables :- A, B, C, (Regions)

Domain :- { Red, Green, Blue }

Constraint :- $A \neq B, B \neq C, C \neq A$

(If all are adjacent)

Conclusion :-

* A solution is achieved when all regions are coloured without violating any constraints.

1b) Define a cryptarithmic problem. solve the cryptarithmic puzzle? $SEND + MORE$

$= MONEY$

A) Cryptarithmic problem

* A cryptarithmic problem is a type of puzzle in artificial intelligence where letters represent digits. The goal is to find a unique digit for each letter such that the arithmetic equation is correct, with no two letters having the same digit.

Given puzzle :-

$SEND + MORE \neq MONEY$

Assign digits to letters as follows:

$S = 9, E = 5, N = 6, O = 7, M = 1, D = 0,$

$R = 8, Y = 2$

(1000000 910 110 00)

Verifying the words as follows :

$$\text{SEND} = 9567$$

$$\text{MORE} = 1085$$

$$\text{MONEY} = 10652$$

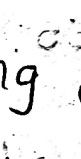
$$\Rightarrow 9567 + 1085 = 10652$$

Conclusion :-

* Each letter has a unique digit and satisfies the given equation.

2) what is adversarial search? Explain

the minmax algorithm using following game tree?

A) 

Adversarial search :-

* Adversarial search is a technique in artificial intelligence used for decision making in competitive environments, where one player tries to maximize the outcome while the opponent tries to minimize it.

Minimax algorithm :-

* The minimax algorithm is used in two

Player games

→ Max player tries to get the highest possible value.

→ Min player tries to get the lowest possible value.

Minimax algorithm using given tree

* In the given tree:

→ upward triangle (Δ) = MAX node

→ Downward triangle (∇) = MIN node.

* The algorithm evaluates from bottom to top process.

step 1:- Evaluate MIN nodes.

choose the minimum from each group:

$$\min(4, 3, 8) = 3 \quad \min(4, 3) = 3$$

$$\min(2, 1) = 1 \quad \min(0) = 0$$

$$\min(4, 2, 3) = 2 \quad \min(2, 8, 4) = 2$$

$$\min(6, 4) = 4 \quad \min(3, 7) = 3$$

$$\min(7) = 7 \quad \min(5, 4, 1) = 1$$

$$\min(5, 2) = 2$$

$$\min(1, 9, 0) = 0$$

step 2 :- Evaluate MAX nodes

choose the maximum from each side :

left side :

$$\max(3, 1) = 3$$

$$\max(2) = 2$$

$$\max(4, 7, 2) = 7$$

Right side :

$$\max(0) = 0$$

$$\max(3, 0) = 3$$

$$\max(2, 3, 1) = 3$$

step 3 :- Evaluate MIN nodes

choose the minimum from each side :

left side :

$$\min(3, 2, 7) = 2$$

Right side :

$$\min(0, 3, 3) = 0$$

step 4 :- Evaluate MAX nodes

choose the maximum from root :

$$\max(2, 0) = 2$$

Final answer :-

The optimal value at the root node is 2.

2a) Explain the resolution algorithm for Propositional Logic?

A)

Resolution algorithm (Propositional Logic) :-

* The Resolution algorithm is a inference method in Artificial Intelligence used to Prove whether a statement is true or false by using Logical clauses.

Steps :-

1. Convert all given statements into conjunctive normal form.
2. Negate the statement that needs to be Proved like.
3. Apply the resolution rule: combine two clauses that contain complementary literals and remove them.
4. Continue resolving until:
 - An empty clause is obtained :- proof is T
 - No new clause can be formed :- Proof is F.

conclusion :-

Resolution works by repeatedly eliminating contradictions and if it produces an empty clause, the given statement is

Logically proven.

3b) show that A can be inferred using the resolution method from the following statements:

i) $(B \wedge C) \rightarrow A$

iv) $D \wedge \neg F$

ii) $(D \wedge E) \rightarrow C$

iv) B

iii) $E \vee F$

A)

Resolution proof :-

* In Artificial intelligence, to prove A using resolution.

step 1 :- Convert to CNF

* $(B \wedge C) \rightarrow A \Rightarrow \neg B \vee \neg C \vee A$

* $(D \wedge E) \rightarrow C \Rightarrow \neg D \vee \neg E \vee C$

* $E \vee F \Rightarrow E \vee F$

* $D \wedge \neg F \Rightarrow D, \neg F$

* $B \Rightarrow B$

step 2 :- List all clauses

$\Rightarrow \neg B \vee \neg C \vee A$

$\Rightarrow \neg D \vee \neg E \vee C$

$$\Rightarrow E \vee F$$

$$\Rightarrow D$$

$$\Rightarrow \neg F$$

$$\Rightarrow B$$

Step 3 :- Apply Resolution

$$E \vee F \text{ and } \neg F \Rightarrow E$$

$$\neg D \vee \neg E \vee C \text{ and } E \Rightarrow \neg E \vee C$$

$$\neg E \vee C \text{ and } E \Rightarrow C$$

$$\neg B \vee \neg C \vee A \text{ and } B \Rightarrow \neg C \vee A$$

$$\neg C \vee A \text{ and } C \Rightarrow A$$

$\therefore$ A is successfully derived, hence

Proved using resolution.

4) Represent the following statements using First order Logic

i) one's mother is one's female parent.

ii) one's husband is one's male spouse.

iii) parent and child are inverse relations

iv) A grandparent is Parent of one's parent.

A)

Let the predicates be :

Parent (x, y) : x is parent of y

child (x, y) : x is child of y

grandparent (x, y) : x is grandparent of y

mother (x, y) : x is mother of y

father (x, y) : x is father of y

husband (x, y) : x is husband of y

spouse (x, y) : x is spouse of y

male (x) , female (x)

First order Logic statements

i) one's mother is one's female parent.

$$\Rightarrow \forall x \forall y [\text{mother}(x, y) \leftrightarrow (\text{Parent}(x, y) \wedge \text{female}(x))]$$

ii) one's husband is one's male spouse.

$$\Rightarrow \forall x \forall y [\text{husband}(x, y) \leftrightarrow (\text{spouse}(x, y) \wedge \text{male}(x))]$$

iii) Parent and child are inverse relations.

$$\Rightarrow \forall x \forall y [\text{Parent}(x, y) \leftrightarrow \text{child}(y, x)]$$

iv) A grandparent is parent of one's parent

$$\Rightarrow \forall x \forall y [\text{Grandparent}(x, y) \leftrightarrow \exists z (\text{parent}(x, z) \wedge \text{parent}(y, z))]$$

∴ These FOL statements formally represent the given relationships using predicates, quantifiers and logical connectives in a clear and precise way.

5a) what is Unification? Explain with suitable examples?

A)

Unification :-

* Unification is a process in artificial intelligence used to make two logical expressions identical by finding suitable substitutions for their variables.

Ex :-

Expression 1: $\text{parent}(x, \text{Kiran})$

Expression 2: $\text{parent}(\text{Padma}, y)$

Unification result :

$x = \text{padma} \quad y = \text{Kiran}$

After substitution $\rightarrow$ Both become

$\text{parent}(\text{padma}, \text{Kiran})$

conclusion:-

Unification helps in matching expressions and widely used in inference and Logic based reasoning.

5b) As per the law, it is a crime for an american to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all missiles were sold to it by Colonel west, who is an american citizen. An enemies of america are considered hostile? prove that "west is a criminal" using Forward chaining.

A)

step 1:- Convert statements into facts and rules.

Facts:

american (west)

enemy (nono, america)

missile (m)

owns (Nono, m)

sells (west, m, Nono)

Rules:-

- * $\forall x, y, z [\text{American}(x) \wedge \text{weapon}(y) \wedge \text{sells}(x, y, z) \wedge \text{Hostile}(z) \rightarrow \text{criminal}(x)]$
- * $\forall x [\text{missile}(x) \rightarrow \text{weapon}(x)]$
- * $\forall x [\text{Enemy}(x, \text{america}) \rightarrow \text{Hostile}(x)]$

Step 2 :- Apply forward chaining

- * from : $\text{missile}(m) \Rightarrow \text{weapon}(m)$
- * from : $\text{Enemy}(\text{Nono}, \text{america}) \Rightarrow \text{Hostile}(\text{nono})$
- * we have : $\text{American}(\text{west}), \text{sells}(\text{west}, m, \text{nono})$
- * Apply the 1st rule : $\text{American}(\text{west}) \wedge \text{weapon}(m) \wedge \text{sells}(\text{west}, m, \text{nono}) \wedge \text{Hostile}(\text{nono}) \Rightarrow \text{criminal}(\text{west})$

$\therefore$ Using forward chaining method, we derive that west is a criminal.

6a) Illustrate forward state space search for planning with a suitable example?

A)

Forward state space search :

* In Artificial Intelligence, forward state space search is a planning method where we start from the initial state and apply step by step until it reached the goal state.

Key idea :-

- Begin with the initial state.
- Apply all possible actions.
- Generate new states.
- Continue expanding states.
- The goal condition is satisfied.

Example (sample plan) :-

Problem : Move from Room A to Room C

⇒ initial state : At A

Goal state : At C

search steps :-

1. start : At A

2. Apply move (A → B) : At B

3. Apply move ($B \rightarrow c$) : At c

4. End : ATCC

Conclusion :- Forward state space search builds the solution step by step from the start, making it easy to understand and implement for planning problems.

6b) Discuss hierarchical planning with an example?

A)

Hierarchical planning :

* Hierarchical planning is a method in AI where a complex problem is solved by breaking it into smaller sub problems arranged in diff. lvs of abstraction.

Key idea :-

- start with a high level goal.
- Decompose it into subtasks.
- continue dividing until we get simple actions.

Example (Travel planning) :-

problem :- Home to college

High Lvl tasks :-

- * Getting ready
- * Travel to college

sub tasks :-

- * Getting ready → wake up & dress
- * Travel to college → take bus / ride bike

Final actions :-

wake up → dress → get ready → take bus
→ reach college.

Conclusion :- Hierarchical planning simplifies complex problems by dividing them into manageable steps, making planning more efficient and structured.

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